

Android Development Practice Set with Sample Answers

1. What is Android?

Sample Answer:

Android is an open-source mobile operating system developed by [Google](#). It is based on the Linux kernel and is used in smartphones, tablets, smart TVs, and wearable devices.

2. Explain the history of Android.

Sample Answer:

Android Inc. was founded in 2003 by Andy Rubin and his team. [Google](#) acquired Android Inc. in 2005, and the first commercial Android phone, the HTC Dream (also called T-Mobile G1), was launched in 2008.

3. What is Open Handset Alliance (OHA)?

Sample Answer:

Open Handset Alliance is a group of technology companies that work together to develop open standards for mobile devices and support the Android platform.

4. List any four features of Android.

Sample Answer:

1. Open-source platform
 2. Multitasking support
 3. Rich multimedia capabilities
 4. Connectivity support such as Wi-Fi and Bluetooth
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5. What are Android devices?

Sample Answer:

Android devices are electronic devices that run Android OS, such as smartphones, tablets, smart watches, televisions, and automobile infotainment systems.

6. What is Android Runtime (ART)?**Sample Answer:**

Android Runtime (ART) is the environment that executes Android applications and improves performance by compiling app code before execution.

7. Draw and explain the architecture of Android.**Sample Answer:**

Android architecture has five layers:

1. Applications
2. Application Framework
3. Android Runtime (ART)
4. Native Libraries
5. Linux Kernel

Each layer provides services to the layer above it.

8. What are the advantages of Android?**Sample Answer:**

- Free and open source
 - Supports many device manufacturers
 - Large number of apps on [Google Play](#)
 - Highly customizable
 - Strong developer community
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9. Compare Android with iOS.

Sample Answer:

Android is open-source and allows extensive customization, while [Apple iOS](#) is closed-source and optimized for Apple devices only.

10. What is AndroidManifest.xml?**Sample Answer:**

AndroidManifest.xml is a configuration file that contains application information such as package name, activities, permissions, and minimum Android version.

11. Explain the Android project directory structure.**Sample Answer:**

Important folders include:

- java/ or kotlin/ – source code
 - res/ – layouts, images, strings
 - manifests/ – AndroidManifest.xml
 - Gradle Scripts – build configuration files
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12. What is the purpose of the res folder?**Sample Answer:**

The res folder stores resources such as XML layouts, images, icons, colors, and string values used in the application.

13. What are Android development tools?**Sample Answer:**

Common tools include:

- [Android Studio](#)
- Android SDK
- Emulator
- ADB (Android Debug Bridge)
- Gradle

14. What is Gradle?**Sample Answer:**

Gradle is a build automation tool used to compile, test, and package Android applications.

15. What is an Android Emulator?**Sample Answer:**

An Android Emulator is a virtual device that allows developers to run and test applications without using a physical Android device.

Unit 2: Views and User Interface

16. What is a View in Android?**Sample Answer:**

A View is the basic building block of the Android user interface. Examples include TextView, Button, and EditText.

17. What is a ViewGroup?**Sample Answer:**

A ViewGroup is a container that holds and arranges multiple Views, such as LinearLayout, RelativeLayout, and ConstraintLayout.

18. What is a Button widget?**Sample Answer:**

A Button is a clickable UI component used to perform an action when the user taps it.

19. What is EditText?**Sample Answer:**

EditText is a widget that allows users to enter and edit text.

20. What is TextView?

Sample Answer:

TextView is used to display read-only text on the screen.

21. What is CheckBox?

Sample Answer:

CheckBox allows users to select one or more options from a list.

22. What is ToggleButton?

Sample Answer:

ToggleButton is a two-state button that switches between ON and OFF states.

23. What are Android layouts?

Sample Answer:

Layouts are containers used to organize UI components on the screen. Examples include LinearLayout, FrameLayout, and ConstraintLayout.

24. What are styles and themes?

Sample Answer:

Styles define the appearance of individual widgets, while themes define the overall look and feel of the entire application.

25. What is the difference between px, dp, and sp?

Sample Answer:

- **px (pixels):** Actual screen pixels
- **dp (density-independent pixels):** Used for layout dimensions
- **sp (scale-independent pixels):** Used for text size and respects user font settings

26. What is screen density in Android?

Sample Answer:

Screen density refers to the number of pixels per inch (PPI) on a device screen. Devices are categorized as ldpi, mdpi, hdpi, xhdpi, xxhdpi, and xxxhdpi.

27. Why is dp preferred over px for layout design?

Sample Answer:

dp provides consistent sizing across devices with different screen densities, while px varies depending on the actual pixel count.

28. How is dp converted to px?

Sample Answer:

Formula:

$$px = dp \times \frac{dpi}{160}$$

For example, 20 dp on a 320 dpi screen becomes 40 px.

29. What is sp and where is it used?

Sample Answer:

sp (scale-independent pixels) is used for text sizes. It adjusts according to both screen density and the user's font-size preference.

30. What is screen orientation?

Sample Answer:

Orientation refers to the direction in which the device is displayed, mainly Portrait and Landscape.

31. What is the purpose of themes in Android?

Sample Answer:

Themes apply a consistent design to all screens in an application, including colors, fonts, and widget styles.

32. Differentiate between styles and themes.

Sample Answer:

A style applies to a single widget, while a theme applies to an entire activity or application.

33. What is ConstraintLayout?**Sample Answer:**

ConstraintLayout is a flexible layout that allows positioning views relative to each other using constraints.

34. What is LinearLayout?**Sample Answer:**

LinearLayout arranges child views in a single row or column.

35. What is FrameLayout?**Sample Answer:**

FrameLayout displays views stacked on top of one another and is commonly used to host fragments.

Unit 3: Activities and Fragments

36. What is an Activity?**Sample Answer:**

An Activity represents a single screen with a user interface, such as a login screen or settings page.

37. Explain the Activity lifecycle.**Sample Answer:**

The main lifecycle methods are onCreate(), onStart(), onResume(), onPause(), onStop(), onRestart(), and onDestroy().

38. What is onCreate() used for?

Sample Answer:

onCreate() is called when the activity is first created and is used to initialize components and set the layout.

39. What is an Intent?**Sample Answer:**

An Intent is a messaging object used to start activities, services, or send broadcasts.

40. What are the types of Intents?**Sample Answer:**

1. Explicit Intent
 2. Implicit Intent
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41. What is an Explicit Intent?**Sample Answer:**

An Explicit Intent directly specifies the target component to be launched.

42. What is an Implicit Intent?**Sample Answer:**

An Implicit Intent requests an action without naming a specific component, allowing Android to choose a suitable app.

43. What is an Intent Filter?**Sample Answer:**

An Intent Filter declares which intents a component can respond to.

44. How is data passed between activities?**Sample Answer:**

Data is passed using Intent extras as key-value pairs.

45. What is a Fragment?

Sample Answer:

A Fragment is a reusable portion of a user interface that can be embedded inside an activity.

46. Why are Fragments useful?

Sample Answer:

Fragments improve modularity and support adaptive layouts for phones and tablets.

47. What are the main Fragment lifecycle methods?

Sample Answer:

onAttach(), onCreate(), onCreateView(), onStart(), onResume(), onPause(), onStop(), and onDestroy().

48. What is the difference between an Activity and a Fragment?

Sample Answer:

An Activity is a complete screen, while a Fragment is a smaller reusable section within an Activity.

49. What is a Service in Android?

Sample Answer:

A Service is a component that performs long-running operations in the background without a user interface.

50. Give one real-life example of a Service.

Sample Answer:

A music player app continues playing songs in the background even when the user opens another application.

51. What is a Broadcast Receiver?

Sample Answer:

A Broadcast Receiver is an Android component that responds to system-wide or

application-wide broadcast messages, such as battery low or network connectivity changes.

52. Give one real-life example of a Broadcast Receiver.

Sample Answer:

An application that displays a warning when the device battery level becomes low uses a Broadcast Receiver.

53. What is a Content Provider?

Sample Answer:

A Content Provider is a component used to share structured data between different applications in a secure manner.

54. Give an example of data accessed through a Content Provider.

Sample Answer:

Contacts stored on the device can be accessed through the Contacts Content Provider.

55. What is the purpose of startActivity()?

Sample Answer:

startActivity() is used to launch a new activity from the current activity.

56. What is startService() used for?

Sample Answer:

startService() is used to start a background service.

57. What is the role of Bundle in Android?

Sample Answer:

A Bundle stores key-value pairs and is commonly used to pass data between activities and fragments.

58. What is the Back Stack?

Sample Answer:

The Back Stack stores activities in the order they are opened, allowing users to return to previous screens using the Back button.

59. What happens when an activity enters onPause()?**Sample Answer:**

The activity loses focus but may still be partially visible. It should save temporary data and release lightweight resources.

60. Why is onDestroy() called?**Sample Answer:**

onDestroy() is called before the activity is removed from memory, allowing final cleanup.

Unit 4: SQLite Database and Device Features

61. What is SQLite?**Sample Answer:**

SQLite is a lightweight relational database built into Android for storing structured data locally.

62. What is a Cursor in Android?**Sample Answer:**

A Cursor is an object used to read and navigate through the results returned by a database query.

63. What are ContentValues?**Sample Answer:**

ContentValues is a class used to store column-value pairs when inserting or updating records in SQLite.

64. Why should a database be closed after use?

Sample Answer:

Closing the database releases resources and prevents memory leaks.

65. What is a table in SQLite?

Sample Answer:

A table is a collection of rows and columns used to store related data.

66. What is a primary key?

Sample Answer:

A primary key uniquely identifies each row in a table.

67. What is the use of SQLiteOpenHelper?

Sample Answer:

SQLiteOpenHelper helps create, upgrade, and manage SQLite databases.

68. What are Android sensors?

Sample Answer:

Sensors are hardware components that detect motion, light, orientation, and environmental conditions.

69. Give examples of sensors used in Android devices.

Sample Answer:

Accelerometer, gyroscope, proximity sensor, and light sensor.

70. What is Bluetooth in Android?

Sample Answer:

Bluetooth enables wireless communication between nearby devices over short distances.

71. What is Geolocation?

Sample Answer:

Geolocation determines the device's physical location using GPS, Wi-Fi, or mobile networks.

72. What is GPS?**Sample Answer:**

Global Positioning System (GPS) uses satellites to provide accurate location information.

73. What is SMS?**Sample Answer:**

SMS (Short Message Service) allows sending and receiving text messages.

74. What is MMS?**Sample Answer:**

MMS (Multimedia Messaging Service) allows sending images, audio, video, and text.

75. Why are graphics and animation used in Android apps?**Sample Answer:**

Graphics and animation improve visual appeal, user engagement, and overall user experience.

76. What is the purpose of the insert() operation in SQLite?**Sample Answer:**

The insert() operation adds a new row of data into a database table.

77. What is the purpose of the update() operation in SQLite?**Sample Answer:**

The update() operation modifies existing records in a table.

78. What is the purpose of the delete() operation in SQLite?

Sample Answer:

The delete() operation removes one or more records from a table.

79. What is the difference between internal storage and SQLite?

Sample Answer:

Internal storage saves files, while SQLite stores structured data in tables.

80. How can sensors improve mobile applications?

Sample Answer:

Sensors enable features such as motion detection, screen rotation, step counting, and automatic brightness control.

81. What are common uses of Bluetooth in Android apps?

Sample Answer:

Bluetooth is used for file transfer, wireless audio, fitness trackers, and smart home device control.

82. Why is geolocation important in mobile applications?

Sample Answer:

It enables navigation, ride-sharing, nearby search, and location-based recommendations.

83. What permissions are typically needed for sending SMS?

Sample Answer:

The application usually requires permission to send and read SMS messages.

84. What is the role of graphics in Android applications?

Sample Answer:

Graphics are used to display icons, images, charts, and custom visual elements.

85. What is the role of animation in Android applications?

Sample Answer:

Animation makes screen transitions and user interactions smoother and more engaging.

Unit 5: Security and Promotion Strategies

86. Why is security important in Android applications?

Sample Answer:

Security protects user data, prevents unauthorized access, and increases user trust.

87. What is a signing certificate?

Sample Answer:

A signing certificate is a digital certificate used to verify the identity of the application developer.

88. Why must Android applications be signed?

Sample Answer:

Every Android application must be signed before installation or publication to ensure authenticity and integrity.

89. What is a keystore?

Sample Answer:

A keystore is a secure file that stores signing keys and certificates.

90. What is Google Play?

Sample Answer:

[Google Play Console](#) is Google's official platform for publishing and managing Android applications.

91. What are the steps to publish an Android application?

Sample Answer:

Prepare a signed APK or App Bundle, create a developer account, upload the app, add store details, and submit for review.

92. What is monetization?

Sample Answer:

Monetization is the process of earning revenue from an application.

93. Name common monetization strategies.

Sample Answer:

Advertisements, paid apps, in-app purchases, and subscriptions.

94. What is application promotion?

Sample Answer:

Application promotion includes activities used to increase downloads and user engagement.

95. What is App Store Optimization (ASO)?

Sample Answer:

ASO improves app visibility using optimized titles, descriptions, keywords, and screenshots.

96. What is [Google Analytics](#)?

Sample Answer:

Google Analytics is a tool used to measure user behavior, engagement, and application performance.

97. How does Google Analytics help developers?

Sample Answer:

It provides information about user sessions, retention, screen views, and conversion rates.

98. What is a crash report?

Sample Answer:

A crash report contains details about application failures to help developers identify and fix issues.

99. Why are user reviews important?

Sample Answer:

User reviews improve credibility, influence downloads, and provide feedback for future improvements.

100. How can a new Android application attract more users?

Sample Answer:

By using ASO, social media marketing, advertisements, positive reviews, and regular updates.